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List of Acronyms

BEM Boundary Elements Method
EPWE Extended Plane Wave Expansion
FEM Finite Elements Method
GEP Generalized Eigenvalue Problem
GM Global Method
HR Helmoltz Resonator
JCA Johnson-Champoux-Allard
MST Multiple Scattering Theory
PEM Poroelastic material
PML Perfectly Matched Layer
PWE Plane Wave Expansion
SAFE Semi-analytical Finite Elements
SCM Spectral Collocation Method
SRR Split-Ring Resonator
TMM Transfer Matrix Method
WFE Wave Finit Elements

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Introduction

Context of the work

Noise is a major issue in our modern societies, ranging from everyday life inconveniences, work comfort problems to an actual pollution problem in severly affected areas. Sources of noise can be diverse; in urban environments, concerns are mostly on traffic and population noise. From industrial origins, noise can have various consequences, causing not only risks to the workers and the neighbouring populations, but also structural hazard to installations. Generally, motorized transport is a major issue in all environments, including aircraft, public transportation, roads, commercial vessel traffic, etc.

The European Environment Agency estimated in 2021 that 1 in 5 European resident are exposed to unhealthy levels of noise, its main cause being traffic. These noise exposures can have a strong impact on health[1] favoring impaired circadian rythm, oxidative stress and cardiovascular dysfunction[2], as well as psychological consequences like anxiety and depression[3]. Moreover, anthropic noise pollution also affects biodiversity at large, causing stress, behaviour changes, and masking sensory perception[4]. Marine traffic was shown to generate stress on marine mammals and modify their behaviour by provoking area avoidance and vocalization disruption[5].

Although sound intensity can be measured objectively, the impact of noise on individuals is in part shaped by social and cultural context. Studies show that expected behaviours norms and customs strongly influence reported annoyance[6], with comparative surveys revealing clear cross-cultural differences. More recent work has linked collectivistic cultural norms to systematically higher everyday noise exposure as an example[7,8]. Through these examples, we can conclude that noise is a very broad topic that raises complex questions in various scientific disciplines, regarding its impact on human life and biodiversity at large. In this work, we want to focus on noise mitigation using engineered structures in acoustics.

Most noise sources have emerged from the modernisation of societies through various innovations; with them come with the need to mitigate their noise impact up to our current capabilities, which is able to evolve thanks to research on noise barriers and acoustic insulation systems. Various strategies have been applied throughout the ages. Dating back to 4th century BC, acoustic jars were used to improve the room acoustics in Greek and Roman theatres as well as in the Middle Ages in Western Europe in churches[9]. These resonators are thought to be capable of reducing reverberation time in buildings, covering a frequency range matching that of human spoken voice. As another example, a famous roman amphitheater was shown to have a seat row disposition able to improve intelligibility of speeches from the stage[10].

Porous materials

Porous and fibrous materials have been traditionally used to absorb sound in the last century[11–16], contemporarily to these antique devices. Largely used in the industry, they are efficient at frequencies starting close to the quarter wavelength resonance frequency of the layer and have been extensively analyzed in the literature[17–21]. They can be found in natural occurences or engineered and cover very large use cases of acoustic insulation such as industry, transport, aerospace, concert halls and many others.

These porous materials are lossy: an airborne acoustic wave propagating through a porous layer is dissipated as it propagates in the bulk, due to mainly two phenomena. Viscous friction occurs due to the drag between the air and the pore walls and another portion of the energy is converted to heat due to the small ducts and subsequently the existence of a thermal boundary layer. These losses mechanisms are strongly dependant on the nature of the material, as they are mainly affected by the geometry of the microstructure. Common models usually account for homogenized effective parameters to characterize these materials as will be discussed in a further section. Since wave velocity in porous media tends to be smaller than that of air, thus lowering slightly the frequency of the quarter-wavelength resonance and coincidentally corresponding to the frequency where such porous structure usually start to have an efficient acoustic absorption. Section 1.1 is dedicated to presenting the various models used for porous and poroelastic media modelling. Their absorption efficiency depends thus on their inherent losses but also on the impedance mismatch between the layer and the medium from which the impinging acoustic wave is originated (i.e usually air). The acoustic impedance mismatch leads to reflection at the interface, while the energy transmitted through the interface in the porous layer gets attenuated. In real-life applications, these materials are not closed systems and there is a need to account for their coupling to surrounding media into which part of the acoustic energy is susceptible to radiate.

A brief history of Acoustic Metamaterials

In more recent years, the emergence of more complex engineered structures has disrupted the landscape of acoustic absorbers. Metamaterials are composite structures designed to exhibit physical effective properties that go beyond the capability of the sole material that composes the bulk. They can be created by a lot of various means, including engineered microstructure, a gradial evolution of the material properties, or repeating patterns or scatterers implemented in one or several directions in space and embedded in a host material as shown for various dimensions in Fig. I.1. Compared to homogeneous layers, they generally lead to more complex structures, and are able to achieve enhanced effects. Usually allowing for a tunable response, these structures offer great flexibility and open new perspectives in material design. This field has been initiated by Veselago, who has theoretically described the electrodynamics of materials having a negative refraction index (occuring in the case of a doubly-negative permittivity and permeability), now known as the Veselago lens in electromagnetic metamaterials[22]. These discoveries have subsequently sparked interest in various disciplines, such as photonics, optics[23] and in the case under consideration in this work, acoustics. Metamaterials can be used to design antennas and absorber for electromagnetic waves and seismic protection devices among others. In acoustic, they allow for various applications such as absorption, filtering, diffusion or cloaking.

Seminal works have been done on the study of the use of lattice arrangements or crystals for wave control applications. Photonic crystals are used to achieve the control of light and electromagnetic waves[24]. Similarly in acoustics, phononic crystals have been developed for the control of elastic and acoustic waves as the 2D (or 3D) periodic distribution of scatterers in a solid host medium[25] as depicted on Fig. I.1. In the case where the host medium is constituted by a fluid, these structures are called sonic crystals[26] and are often studied through their so-called band structure, also called dispersion diagram/relation when applied to the study of non-periodic geometries, which will be presented in Section **??** and extensively detailed Section 1.5. In the remaining of the literature overview, focus will be centered around metamaterials that involve periodicity in only one direction in most cases. These acoustic metamaterials can be divided into two main categories: the first concerns the so-called *reflection problem* where the case study is that of an incoming wave on some sample or panel which has a rigid-backing at the end. There, the consideration is focused on mitigating the reflected wave and acheving important acoustic absorption. The other case is that of a *transmission problem* where the incoming wave is able to be transmitted through the sample. In this case and depending on the application, either or both reflection and transmission are sought to be mitigated. In each of these cases respectively, the structures can be called metasurfaces and interfaces.

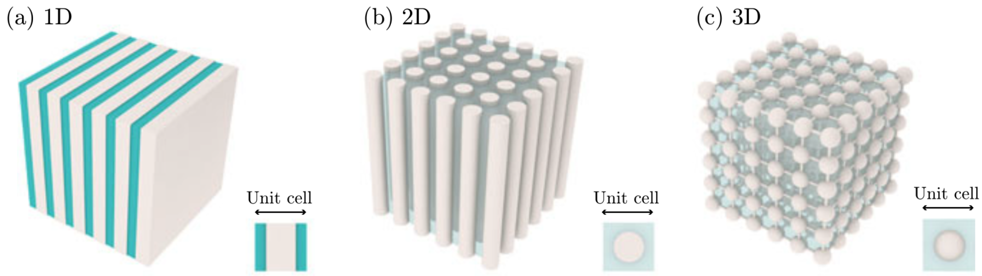

Fig. I.1: Periodic cells in 1, 2 and 3D. Image taken from Ref.[27]

Structured materials that tackle the reflection problem can involve many various strategies. The combination of a perforated plate and coiled-up channels is able to achieve large broad-band absorption underwater[28]. Micro-perforated plate can also be used to achieve interesting absorption properties for airborne acoustics[29]. Materials structured from resonant building blocks with resonance frequencies in the sub-wavelength regime i.e., for frequencies which wavelength is much bigger than the typical size of the material, have also been studied[30–32]. In order to achieve these effects, such systems usually profit from the effects of bandgaps, which are frequency bands in the band structure where wave propagation is forbidden due to the destructive interference due to the presence of scatterers, have been achieved using locally resonant cells in the design of a sonic crystal[33]. These resonances introduce strong dispersion of the waves and the properties of the system are thus dramatically affected. Together with the unavoidable losses, the resonant behaviour of metamaterials make them perfect candidates to design efficient absorbers at low frequencies with reduced dimensions[34,35]. Balancing the dissipation and leakage of each resonance yields the recently described *critical coupling*. In this case, the reflection drops to zero[36]. Perfect absorbers can thus be designed for reflection problems for both monochromatic waves[34,37] and broad range of frequencies[37–41]. This can be achieved by tuning the structure using the scattering matrix zeros plotted in the complex frequency plane; critical coupling would be achieved when the scattering matrix zero has a vanishing imaginary frequency. Similarly to these broadband absorbers, Yang *et al.* have realized the backing of a porous layer using Fabry-Pérot channels to design an efficient structure[42].

One of the initial interests of the acoustic metamaterials community was the possibility of producing deeps in the transmission spectra at very low frequencies. This has been widely developed in the literature, a small subset of these works include Refs.[43–46]. These configurations are based on resonant elements and display a zero transmission in the lossless case at their resonance frequency. In reality, losses cannot be avoided and a small amount of the energy is always transmitted. Membrane-type acoustic metamaterials[47–49], or acoustic metamaterials made by Helmholtz resonators[50–52] have been used to develop systems with high values of transmission loss for audible acoustics. Considering the vibroacoustic coupling allows for more precise depiction of real-life applications. Works on periodic soft inclusion in a soft solid matrix was done by Aberkane *et al.*[53] showing improved transmission loss affected by the presence of soft inclusions in the solid layer. In this case, the presence of losses was able to improve the transmission loss at low frequencies. Also accounting for vibroacoustic coupling, attached resonators to walls have been also used to reduce the transmission loss of panels[54,55].

Mirror-symmetric structures are geometries that present a symmetry in the plane of the wave propagation meaning that a wave traveling from the right-side will be affected identically as if propagating from the left-side of the geometry. Their behaviour can be described using a two-port scattering process[56]. It is worth noting here that if only a monopolar resonator (Helmholtz) or a dipolar resonator (membrane) is used to absorb waves, then only 50% of energy can be absorbed as shown in several contributions[56,57]. However, using a material made of slits loaded by an array of identical resonators creates an accumulation point close to the resonance frequency of the resonators allowing the accumulation of the resonances of the slit, thus placing together symmetric and antisymmetric resonances[47,50,58], although requiring for a compromise between absorption and thickness. A more complex approach consists in designing degenerate resonators. By using combinations of membrane and cavity resonators[47,59] and with slits loaded with Helmholtz resonators[57], quasi-total absorption can be reached. In these systems, each resonator manages half of the problem, absorbing each one half of the energy and allowing a perfect absorption peak: all the energy that impinges on the system is dissipated within the system, i.e., the energy is neither reflected nor transmitted. The other alternative to obtain perfect absorption consists in breaking the symmetry of the system. These systems can achieve unidirectional perfect absorption, i.e., perfect absorption can be obtained from one of the two incidence sides[56]. The principle of these systems consists in using two resonators, one that is used to reduce as much as possible the transmission and the other that impedance matches the system from one side[34,37,49]. This concept has been exploited to develop broadband perfect absorbers using a cascade of the previous process giving rise to a rainbow absorber[38].

From porous layers to metaporous surfaces

In order to increase the capabilities of acoustic absorption of porous materials, some works can involve the modification of the core structure of porous and poroelastic materials by tuning of the microstructure, introducing a gradient of the properties, or by using anisotropic materials. With the increase of new fabrication techniques based on 3D printing, the structure of the porous materials can be controlled and tuned to design materials with better absorption capabilities at low frequencies, based on gradient properties[60,61], or on anisotropic materials[62–65]. Similarly, resonators filled with 3D printed porous materials have been developed to design perfect and broadband absorption[66]. Other configurations have been studied, among them the periodic partitioning of the metaporous layer[67] or a labyrinthine channel embedded in a porous host[68], based on the idea of coiling up the space of the channel as was done for non-porous metamaterial designs. Recently, a hierarchical multilayer metaporous structure was proposed using an optimization procedure, giving rise to gradient and periodic designs[69].

Some studies involving porous layers embedding periodic arrays of inclusions were done during the last 15 years[70–73]. These inclusions can have either resonances or not and they allow for an increased dissipation in the viscous regime. These structures are refered to as metaporous surfaces when used in reflection problems. The first works involving metaporous surfaces or interfaces were developed for transmission problems[74,75]. In case of more than one inclusion per spatial period, several absorption peaks can be obtained for a particular arrangement along the porous thickness[71]. Nevertheless, this rapidly leads to a thick structure. Near the inclusion resonance frequency, chosen to be in the viscous regime, an absorption peak occurs[76]. The resonance frequency of the inclusions depends on the orientation of the resonators relative to the rigid boundary[72,76]. Recently, 3D printed porous materials with embedded split-ring resonators have been modeled[64], designed and experimentally characterized showing perfect absorption peaks. More complex spiral-shaped slits have been embedded in a porous layer and have shown to be able to maintain a high absorption for large incidence angles[77]. Xiong *et al.* have studied a metaporous liner with embedded resonator and shown the existence of an exceptional point when the geometry is tuned properly, leading to enhanced absorption[78].

Previous metaporous layers studies have considered a host medium modelled as a rigid-frame porous layer, meaning that the layer is modeled as a lossy fluid if the elastic phase can be assumed motionless; extensions of these structures to consider poroelastic materials, that is considering the skeleton motion coupled to the fluid phase, have been done[82]. Interesting properties were exhibited for thin elastic shell inclusions, and particularly in the case of a coupling between the trapped and the first volume mode leading to a high wide band absorption. While this previous study implements a semi-analytical model that makes use of Multiple Scattering Theory (MST), it is worth noting here that different modelings exists as well, based on a Finite Elements Method (FEM) and MST have been validated for the case of an air-saturated metaporoelastic configuration with a solid inclusion and rubber coatings[83]. This thesis is especially focused on the computation of the dispersion relation of such configurations. Therefore, a dedicated literature review dealing with the tools available to achieve such computation will be given in Chapter 1.

Wave propagation modelling and dispersion analysis

The analysis of wave propagation in periodic and complex media often relies heavily on numerical tools. Two main topics are of interest in this thesis: the scattering of an incident

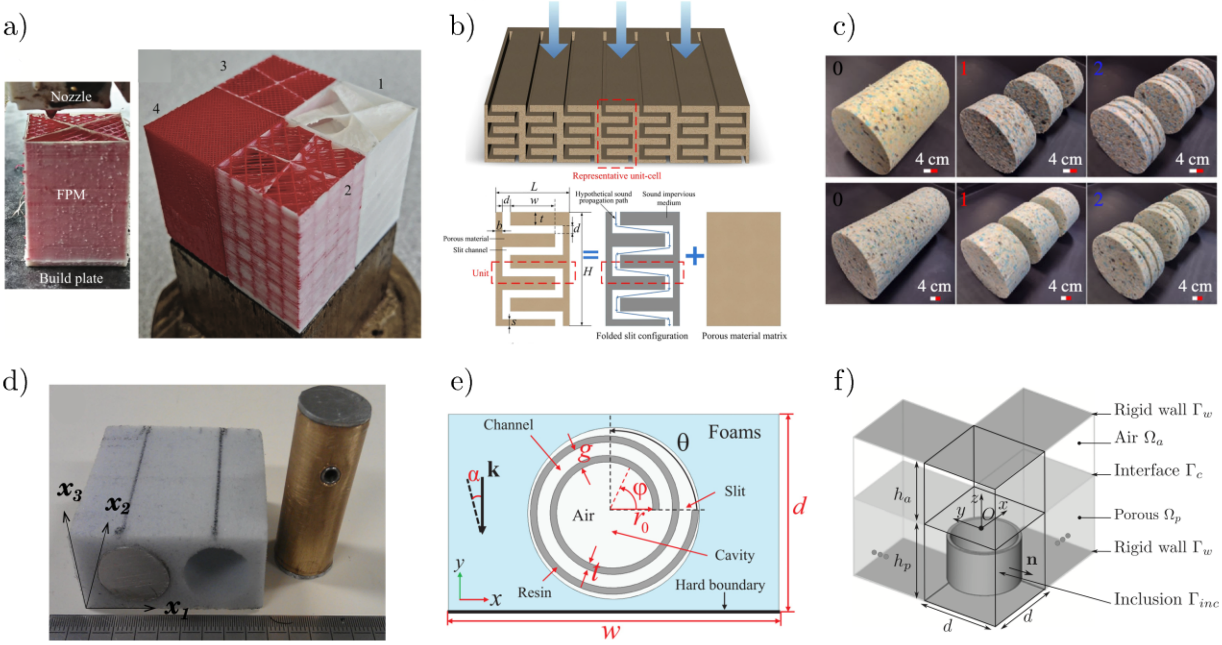

Fig. I.2: Example of metaporous structures: a) 3D printed microstructure with helocoidal shape[79], b) embedded labyrinthine channel[80], c) hierarchical configuration[69], d) Helmholtz resonators[76] e) archimedean spiral[81] f) resonator tuned to an exceptional points[78]

wave by a structure as well as its dispersion relation. The first one is simply the acoustic response of a system, where an incident wave propagates towards a specific geometry and is scattered by the latter. The scattering properties are of utmost importance, since tuning those remains one of the main topics for metamaterial applications[27,31]and the goal is often to reach exceptional properties, such as perfect absorption of the incident wave by the structure. The scattering properties depend on the nature of the material(s) used, on the geometry and generally, the modal behaviour of the whole system.

In order to understand this behaviour in more details, we must turn to the study of the dispersion of the waves. Any causal system undergoing a scattering process can be described through their dispersion equation[84] as an algebraic expression. This allows for the characterisation of the dynamics of the modes in the system. Classically for dielectrics, the complex refractive index is expressed as a function of the frequency[85]. In acoustics, we study dispersion in systems in the form $k(\omega)$, informing on the possible wavenumbers at which the system can respond for a given frequency. Reciprocally, the problem can also be written in the form $\omega(k)$, which is the other way around. From these equations, we can represent dispersion relations, diagrams that represents the wavenumber in a given direction against the frequency[86], thus informing on the evolution of the velocity of the propagation of waves and thus the dispersion and/or attenuation effects that the structure has. With these diagrams, the notions of phase velocity $v_\phi = \omega/k$ to qualify the local velocity of a wave and the group velocity $v_g = \partial\omega/\partial\mathrm{Re}(k)$, defining the velocity of the envelope of a wavepacket or the velocity of the energy. In non-dispersive media, these quantities are the same, while the introduction of dispersion causes a distorsion of a wavepacket with propagation[87].

In the context of metamaterials and periodic structures, dispersion relations are often refered to as band structure and have been widely used to characterize[88] sonic crystals[89]. A theoretical definition of a dispersion relation consists in the spectrum of all solutions of a set of equations given a set of parameters. shows the possible existing states of a system. Generally the form of the dispersion relation, and the various modes that are uncovered can give very in-depth details about the modal behaviour of any structure. As an example, in ultrasonics and non-destructive testing, localizing zero-group velocity (ZGV) points in the dispersion relation is very helpful for a number of uses including defects localization and material characterization[90].

Scarce literature focuses on the computation of the dispersion relations of waves propagating in dissipative layers with open domains general exists: the dispersion of waves in a single poroelastic layer has been studied by Boeckx *et al.*[91,92] while an unpublished paper from Kiefer *et al.* took interest in Lamb waves in viscoelastic layer coupled to surrounding half-spaces[93]. There is an interest in analogous studied of the coupling between surrounding media and metaporous surfaces, as part of the acoustic energy of the wave propagating in the surface should radiate back into the domain, leading to more complex loss effects on the structure.

These aspects will be discussed in detail in the following chapter, both from the theoretical perspective of porous and poroelastic media (and their coupling to surrounding media) and in terms of the available numerical tools for their computation.

Motivations and Contents of the thesis

This thesis is dedicated to the numerical modeling of wave propagation in open-system structures comprised of stacks of layers, homogeneous or with embedded scatterers. As it has been seen in the previous section, these materials combine the dissipative characteristics of porous media with the geometric or structural features typical of acoustic metamaterials, thus offering advanced control over their scattering properties. Understanding and predicting the acoustic behavior of such systems is therefore essential for designing efficient structures. In particular, this work focuses in part on the calculation of the dispersion of waves in these structures, and understand how the existing propagating and evanescent modes are shaping the scattering properties. Therefore, the development of efficient numerical strategies is critical to this study.

The tuning of metamaterials could be achieved in some studies by looking at the zeros of the reflection coefficient in the complex plane, thus highlighting the zeros of the scattering matrix. However, the wavenumber might be able to give out additional informations regarding the acoustic behaviour of metaporous layers. Band structure gives useful informations concerning sonic and phononic crystals. However, the dispersion relation of a metaporous layer embedded in an open system has never been addressed; both mechanisms of energy leakage in the surrounding media and dissipations in the porous layer should be investigated. However, this imply a heavy modification of the interface relations of the structure. It expected to have significant consequences on the propagation of modes therein. This work will be done on both homogeneous layers as well as periodic structures.

From these observations, we should review existing numerical methods able to compute the acoustic response of metaporous structures and consider ways to obtain a precise and robust method able to predict the scattering properties and acoustic responses of metaporous surface and interfaces. Additionally, we should evaluate methods that have been implemented for the computation of guided waves in layers and stacks of layers coupled to semi-infinite media.

Confronting dispersion relations against absorption coefficients should help to highlight the modal behaviour responsible for the various absorption peaks obtained by previously studied configurations and also highlight the full modes existing in the layers which might uncover additional phenomena.

This introductory section has set the context as well as an overview of the literature for the present thesis. We saw that acoustic metamaterials have emerged in the last decades and propose noise mitigation structures that are much more efficient that the traditional foams used for sound insulation. In particular, metaporous surfaces have shown to be promising options achieving one or multiple unitary absorption peaks at subwavelength frequency. Overall, a focus area has been identified towards identifying more precisely the modal behaviour of these configurations. Thus, this thesis focuses on two objectives: the development of numerical methods targeted to the specificities of metaporous surfaces and also on the computation of the dispersion relations of guided waves in these structures.

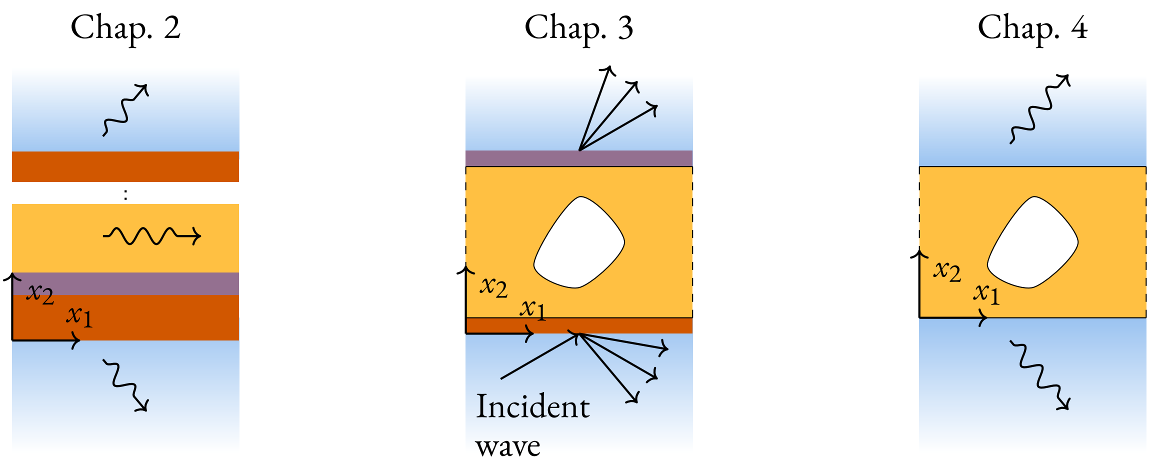

Fig. I.3: Diagram presenting a general layout for the configurations that will be studied in each chapter. The ⇝ arrows represent radiative solutions while the ⟶ arrows represent scattered amplitudes.

Organisation

The manuscript is split into four different chapters as follows,

— CHAPTER 1 introduces the fundamental concepts necessary for the good understanding of the thesis. This chapter focuses on the scattering of waves on dissipative layers and the guided waves formalism that is used for the calculation of dispersion relations. Additional bibliography is proposed on these topics as well as a summary of usable numerical methods to compute such problems. The chapter first introduces notions on the assumptions behind the widespread models of porous and poroelastic media. Next, wave propagation in systems made out of homogeneous layers is studied in the presence of an incident wave and the dispersion problem is studied, that is the free solution of the system of equations, without any forcing. The next sections deal with extension of the two scattering and dispersion solving for periodic unit cells that have embedded scatterers. Finally, a summary of the overviewed methods is proposed with perspectives on the possibilities of our work.

— Chapter 2 targets the study of the propagation of guided waves in multilayer arbitrary structures with homogeneous dissipative layers. A Spectral Collocation Method (SCM) is implemented and extra steps allow for the local coupling of surrounding media allows for the computation of the dispersion relation for these configurations, along with the mode shapes and energy indicators. The formalism is presented in a general manner, so as to allow for any stack of elastic, fluid and poroelastic layers can be described with this method. Results are shown for a bilayer melamine - aluminium system accounting for the surrounding air, compared with a semi-analytical result and include the dispersion relation of the system, the energy flow at given location and the evaluation of the energy transport velocity for some of the branches. An experimental comparison is also proposed. This chapter is constituted from an article that has been published in Journal of Applied Physics[0].

— Chapter 3 is focused on several numerical strategies for the modelling of scattering properties of metaporous/metaporoelastic structures. A plane wave approach is coupled to a FEM via a state vector formalism. Results focused on the performance of the proposed method, compared to other similar frameworks is proposed. The convergence of this method is shown for various metaporoelastic surfaces, which geometries were studied in previous papers. This chapter is a manuscript submitted to International Journal for Numerical Methods in Engineering[0].

— In Chapter 4, the modal behaviour of metaporous structure is studied. The FEM method from the previous chapter is adapted to compute dispersion relations. Results are presented for the case of a rigid inclusion, as well as with embedded Split-Ring Resonator (SRR). Contents for this chapter are part of a draft that will be submitted for publication in Physical Review B.

Chapter 1

Analytical and numerical modelling of wave propagation in dissipative and periodic open systems

Contents

This chapter lays the foundation for the thesis by providing the fundamental theoretical notions and methods used in this work, in order to guide the reader through the various chapters that will follow. At first, an overview of the acoustic modelling of porous media is proposed. Following the presentation of the state of art in the introduction and of the areas of focus that have been identified from the latter, the key notions and methods for the computation of the scattering of waves by metaporous structures will be presented. From there, the formalism for the calculation of dispersion curves will be presented along with an overview of the methods used for such computations. Both of these topics will be illustrated with preliminary results in simple configurations.

1.1 Modelling of porous media

Porous media is a category of materials which encompasses naturally occuring substances such as rocks, soils and biological tissues, and engineered materials like ceramics, metallic foams, rockwool or certain types of polymers. They are considered biphasic, that is constituted of two distinct domains: a pore network Ω_f filled by a saturating fluid embedded in an elastic matrix Ω_s. A diagram of this microscopic structure is shown in Fig. 1.1. The porosity ϕ is defined as the fraction of fluid over the total volume of the material. In the limit case, $\phi = 0$ the material becomes purely a solid while at $\phi = 1$, it becomes a fluid. At microscopic scales, the geometry of the fluid phase thus forms a complex microstructure. Fortunately, at airborne acoustic scales, these materials can often be considered at a meso-/macroscopic level and therefore models have been developed to describe the large-scale motion of these materials. In this context, they can be regarded as homogeneous and described by a set of effective parameters that accounts for their complex geometry. The porous structure of these materials causes the incoming acoustic energy to be dissipated through thermal exchanges and friction (viscous losses) along the duct/pore network, while elastic dissipation take place when the skeleton and the fluid phase are coupled. These losses are thus dependant on the geometry, which parametrized by the porosity, the saturating fluid properties, the pore network size, connectivity and shape, as well as the frame mechanical parameters.

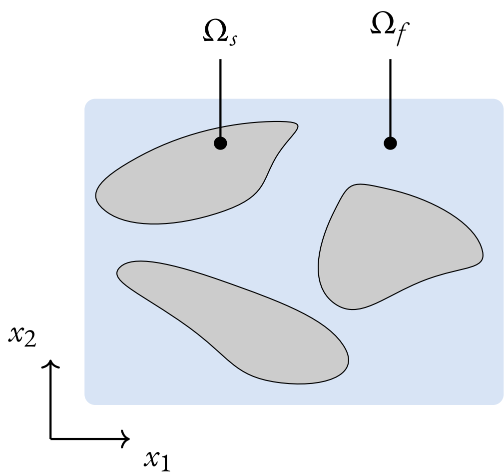

Fig. 1.1: Porous material structure

1.1.1. Rigid-frame approximation

In the so-called rigid-frame approximation, the skeleton of the porous material is considered motionless. Therefore, the pore walls are rigid walls and the material can be modeled as an *equivalent fluid* which possesses complex effective properties in order to model the losses mechanisms. A harmonic acoustic propagation is assumed with a positive time convention $e^{i\omega t}$ troughout this manuscript (with the exception of Chapter ??). Acoustic propagation is governed by,

$$\nabla \cdot \left(\frac{1}{\rho_{eq}(\omega)} \nabla p(\boldsymbol{x}, \omega) \right) = \frac{1}{K_{eq}(\omega)} \omega^2 p(\boldsymbol{x}, \omega) \tag{1.1}$$

where p is the pressure field, ρ_{eq} is the effective density and K_{eq} is the effective bulk modulus. In our case (that of an isotropic and homogeneous medium), this equation can also simply be written as an Helmholtz equation as $(\nabla^2 + k_{eq}^2)p(\boldsymbol{x}, \omega) = 0$ with $k_{eq} = \omega\sqrt{\rho_{eq}/K_{eq}}$ the equivalent fluid wavenumber.

The effective medium parameters are usually calculated using the Johnson-Champoux-Allard (JCA) semi-phenomenological model[94,95], developed from from 1987 to 1991, that accounts for the viscous and inertial dissipative effects. The materials parameters are thus complex and frequency-dependant; the density accounts thus for the viscous effects while the bulk modulus accounts for thermal exchanges. These two quantities are given by the JCA model as,

$$\begin{aligned}
\rho_{\mathrm{eq}}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi} \left(1 + \frac{\mathrm{i} R_f \phi}{\rho_f \alpha_\infty \omega} \sqrt{1 - \mathrm{i}\omega \rho_f \mu_f \left(\frac{2\alpha_\infty}{R_f \phi \Lambda} \right)^2} \right), \\
K_{\mathrm{eq}}(\omega) &= \frac{\kappa P_0}{\phi} \left(\kappa - (\kappa - 1) \middle/ \left(1 + \frac{\mathrm{i} R_f \phi}{\rho_f \alpha_\infty \mathrm{Pr}\omega} \sqrt{1 - \mathrm{i}\mathrm{Pr}\omega \rho_f \mu_f \left(\frac{2\alpha_\infty}{R_f \phi \Lambda'} \right)^2} \right) \right)^{-1}.
\end{aligned} \tag{1.2}$$

These expressions involve a large number of effective parameters for the porous material; α_∞ is the tortuosity, R_f is the flow resistivity, Λ and Λ' are respectively the viscous and thermal characteristic length. The saturating fluid is characterized in this model by its heat capacity ratio κ (commonly also denoted γ), density ρ_f, kinematic viscosity μ_f, pressure P_0 and Prandtl number $\mathrm{Pr} = c_p \mu_f / k$ with c_p is the specific heat and k the thermal conductivity. We can introduce also the Biot frequency $\omega_c = R_f \phi / (\rho_f \alpha_\infty)$ that defines the transition frequency between the inertial and viscous regimes.

1.1.2. Elastic motion of the skeleton: Biot theory

While the equivalent fluid formalism is able to accurately model porous material in specific sets of conditions, that is when the skeleton can be assumed to be motionless, this assumption can quickly become limited as soon as the said skeleton is subjected to an elastic deformation. This is can be the case when the saturating fluid of the PEM has a high density (e.g. water), when the skeleton is coupled to elastic plates or when the acoustic radiation has a wavelength comparable to the pore scale, although these are just examples. In this situation, the full elastic motion should be modeled, so that to account for the coupling between the solid matrix and its fluid filling. The Biot theory[96–99]is a formalism that encompasses the theory of the deformation of porous material saturated by an incompressible fluid, used both in geophysics and acoustics. For sake of conciseness, the various models will be briefly be presented along with the relevant physical quantities. A full derivation would be out of scope and is presented in existing works[11].

Before presenting the full Biot model, some quantities should be provided additionally to the equivalent fluid parameters. The frame is assumed to undergo elastic deformations, thus the common mechanical parameters used are the solid density of the frame material ρ_s as well as its apparent density $\rho_1 = (1-\phi)\rho_s$. The bulk modulii are denoted K_b and K_s, respectively for the frame and the material. The shear modulus is also considered complex, accounting for structural losses as $\tilde{N} = N(1+\mathrm{i}\eta)$ with η the loss factor. Symbols written with a tilde denote complex expressions and therefore account for some sort of loss mechanisms. For the fluid, the apparent density is $\rho_2 = \phi\rho_f$ and we define $\rho_{12} = -\phi\rho_f(\alpha_\infty - 1)$. A summary on Biot theory is also proposed in Chap. ??) with the tilde symbols omitted for ease of reading.

Formulations for the motion equations

The Biot poroelastic equations from the original 1956 formulation[96,98,99] depend on the solid displacement vector $\boldsymbol{u}_s$ and the fluid displacement $\boldsymbol{u}_f$,

$$\begin{aligned} \nabla\cdot\boldsymbol{\sigma}_s(\boldsymbol{u}_s,\boldsymbol{u}_f) &= -\omega^2\tilde{\rho}_{11}\boldsymbol{u}_s - \omega^2\tilde{\rho}_{12}\boldsymbol{u}_f, \\ \nabla\cdot\boldsymbol{\sigma}_f(\boldsymbol{u}_s,\boldsymbol{u}_f) &= -\omega^2\tilde{\rho}_{12}\boldsymbol{u}_s - \omega^2\tilde{\rho}_{22}\boldsymbol{u}_f, \end{aligned} \tag{1.3}$$

where $\boldsymbol{\sigma}^s$ is the partial solid stress tensor, $\boldsymbol{\sigma}^f$ is the partial fluid stress tensor and the apparent densities of the solid-phase is $\tilde{\rho}_{11}$, that of the fluid is $\tilde{\rho}_{22}$ and the coupling apparent density is $\tilde{\rho}_{12}$. They are complex quantities expression that also account for the contribution of viscous forces and are given as,

$$\tilde{\rho}_{22} = \phi^2\tilde{\rho}_{eq}, \quad \tilde{\rho}_{12} = \rho_2 - \tilde{\rho}_{22}, \quad \tilde{\rho}_{11} = \rho_1 - \tilde{\rho}_{12}. \tag{1.4}$$

These coupled motion equations are associated to the stress-strain relation,

$$\begin{aligned} \boldsymbol{\sigma}_{s,ij} &= \left[(\tilde{P}-2\tilde{N})\nabla\cdot\boldsymbol{u}_s + \tilde{Q}\nabla\cdot\boldsymbol{u}_f\right]\delta_{ij} + 2\tilde{N}\boldsymbol{\epsilon}_{s,ij}, \\ \boldsymbol{\sigma}_{f,ij} &= \left(\tilde{Q}\nabla\cdot\boldsymbol{u}^s + \tilde{R}\nabla\cdot\boldsymbol{u}_f\right)\delta_{ij} = -\phi p\delta_{ij}, \end{aligned} \tag{1.5}$$

where $\nabla\cdot\boldsymbol{u}_s$ and $\nabla\cdot\boldsymbol{u}^f$ correspond to the dilatation of the solid and fluid phases respectively and $\boldsymbol{\epsilon}_s$, is the solid-phase strain tensor. The Biot coefficients are denoted as $\tilde{P}$ for the solid elastic coefficient, $\tilde{Q}$ for the elastic-fluid coupling coefficient, and $\tilde{R}$ for the fluid-phase coefficient. In the context of a PEM saturated by a light fluid, their expressions are

$$\tilde{P} = K_b + \frac{4}{3}N + (1-\phi)^2\frac{\tilde{K}_{eq}}{\phi}, \quad \tilde{Q} = \tilde{K}_{eq}(1-\phi), \quad \tilde{R} = \phi\tilde{K}_{eq}. \tag{1.6}$$

Generalized forms of these coefficients are required when the limit $K_b/K_s \to 0$ no longer holds, as is the case for water-saturated PEM, and are given in Ref.[11].

A different formulation, especially convenient for finite-elements discretization is known as the $(\boldsymbol{u}^s, p)$ formulation. First, new parameters in terms of the Biot elastic coefficients and apparent densities can be defined,

$$\hat{A} = \tilde{P} - 2\tilde{N}_s - \frac{\tilde{Q}^2}{\tilde{R}}, \quad \tilde{\rho} = \tilde{\rho}_{11} - \frac{\tilde{\rho}_{12}{}^2}{\tilde{\rho}_{22}}, \quad \tilde{\gamma} = \phi\left(\frac{\tilde{\rho}_{12}}{\tilde{\rho}_{22}} - \frac{\tilde{Q}}{\tilde{R}}\right). \tag{1.7}$$

γ is another coupling coefficient, also denoted as $\tilde{\beta}$ in the derivations of Chapter ??. Using the second row of Eq. (1.3) allows for the expression of the fluid displacement as, $\boldsymbol{u}^f$ =

$\phi/(\tilde{\rho}_{22}\omega^2)\nabla p - \tilde{\rho}_{12}/\tilde{\rho}_{22}\boldsymbol{u}^s$ and using that in Eq. (??) and Eq. (1.5) allow to rewrite the coupled motion equations. Using these, the so-called $(\boldsymbol{u}_s, p)$ formulation of the Biot theory[100] is expressed as,

$$\begin{aligned} \nabla \cdot \hat{\boldsymbol{\sigma}}_s + \tilde{\rho}\omega^2 \boldsymbol{u}_s + \gamma \boldsymbol{\nabla} p = 0, \\ \frac{\nabla^2 p}{\rho_{22}\omega^2} - \frac{\gamma}{\phi^2}\nabla \cdot \boldsymbol{u}_s + \frac{1}{\tilde{R}} p = 0, \end{aligned} \tag{1.8}$$

where $\hat{\boldsymbol{\sigma}}_s$ is the *in vacuo* solid stress tensor,

$$\hat{\boldsymbol{\sigma}}_{s,ij} = \boldsymbol{\sigma}_{s,ij} - \phi \frac{\tilde{Q}}{\tilde{R}} p \delta_{ij} = \hat{A} \nabla \cdot \boldsymbol{u}_s \delta_{ij} + 2\tilde{N} \boldsymbol{\epsilon}_s. \tag{1.9}$$

Propagation in unbounded poroelastic media

Poroelastic media modelling supports three waves: a compression φ_p and shear wave ψ_s related to the solid phase as well as an additional compression wave φ_a, related to the fluid phase. In the case of each of these formulations, the fields can be expressed depending on these potentials,

$$\boldsymbol{u}_s = \nabla \varphi_p + \nabla \varphi_a + \nabla \times \psi_s, \qquad \boldsymbol{u}_f = \mu_p \nabla \varphi_p + \mu_a \nabla \varphi_a + \mu_s \nabla \times \psi_s, \tag{1.10}$$

with μ_p, μ_a, μ_s are coefficients determined by the ratio of solid over fluid displacements ratio. The wavenumbers for each of these waves are, following section 6.5 of Allard and Atalla textbook[11] and denoted as δ_i, $i = 1, 2, 3$ in the text,

$$\begin{aligned} k_{p,a} &= \sqrt{\frac{1}{2}\left(\delta_2^2 + \delta_{eq}^2 \pm \delta_t\right)}, \\ k_s &= \omega \sqrt{\frac{\hat{\rho}}{\tilde{N}}}, \end{aligned} \tag{1.11}$$

respectively for the solid compression, fluid compression and shear wave, with,

$$\delta_{eq} = \omega \sqrt{\frac{\tilde{\rho}_{eq}}{\tilde{K}_{eq}}}, \quad \delta_1 = \omega \sqrt{\frac{\hat{\rho}}{\hat{P}}}, \quad \delta_2 = \omega \sqrt{\frac{\hat{\rho}_s}{\hat{P}}}, \quad \delta_t = \sqrt{(\delta_2^2 + \delta_{eq}^2)^2 - 4\delta_{eq}^2 \delta_1^2}. \tag{1.12}$$

The propagation characteristics, i.e. wavenumbers of the waves, of porous and poroelastic media are illustrated in Figure 1.2. As discussed in the previous sections, the wavenumber for the equivalent fluid as well as the three for the poroelastic layer are complex-valued. Their real part account for the propagation of the phase flow, inversly proportional to the wave velocity and their imaginary part accounts for the attenuation that occurs due to the losses inherent to each of these materials. It appears that at low frequencies, the attenuation is very low for all waves. The velocity of the compression and shear waves (observed on the figures by the slope of the real parts of the wavenumbers) is much higher than the waves in the fluid for the equivalent fluid and the compression fluid phase velocity, as they propagate in a soft solid where the velocity is usually much higher than in a fluid.

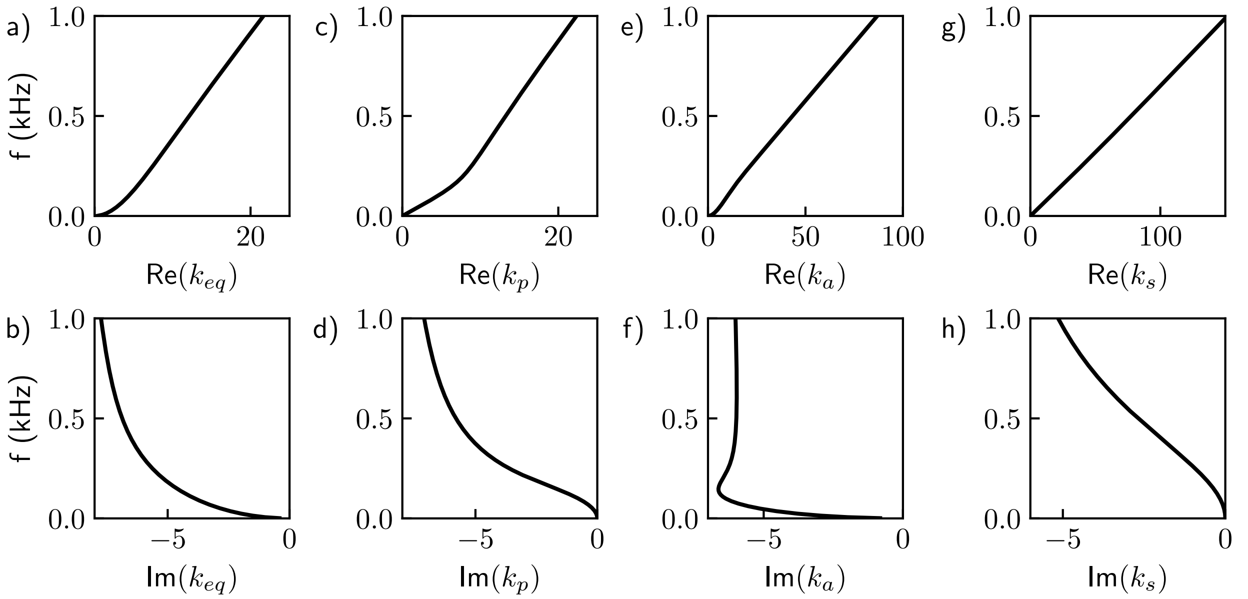

Fig. 1.2: Wavenumbers of waves in unbounded porous media in the case of an a-b) equivalent fluid and c-h) for the Biot model.

1.2 Wave propagation in homogeneous layers

In the previous section, a formulation for rigid-frame porous media and two formulations for poroelastic media modelling have been presented. Concluding the presentation of these motion equations, some characteristics of the wave propagation in unbounded porous and poroelastic media have been shown. The present section proposes the introduction of a geometry, that is to say boundaries or interface conditions delimiting different domains characterized each by a different type of material. Generally, the configurations studied in this work assume a monochromatic plane wave originated from a fluid half-space, i.e. air, impinging at the surface of a structure and is then scattered (reflected and transmitted) from and through each interface. This is illustrated in Fig. 1.3 for the two main cases that are the so-called reflection problem where the structure is bounded on one side by a rigid-backing and the transmission problem where the incident wave is reflected and transmitted through the layer.

If the material in which the wave propagates is bounded by two parallel, free, and flat surfaces, i.e., a plate, then it is refered to as a layer. The waves supported by the medium propagate by reflecting alternatively from one surface to the other; they are said to be guided along the layer. These observations of successive reflections and refractions lead to the concept of waveguides, but their analytical descriptions often leads to writing complex system that can be difficult to solve. To determine the modes of a structure, one must seek to solve the propagation equations and the boundary conditions imposed at its boundaries. In a first section, the scattering of a plane wave by a homogeneous porous layer is investigated along with the various method available. Next, the methods for the computation of the modes of these configurations are presented. In a subsequent section, more complex geometries, those of metaporous structures, are of interest and both methods for the computation of the scattering coefficients and the modes of those configurations will be discussed.

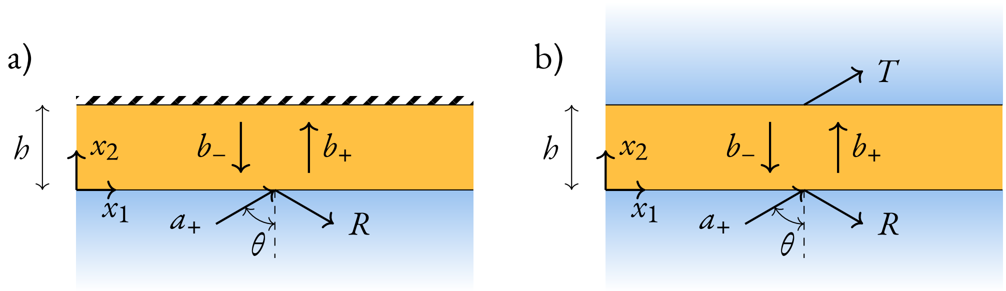

Fig. 1.3: Diagram representing the cases of the a) reflection and b) transmission problem

1.2.1. Analytical plane wave modelling

Modeling acoustic propagation in homogeneous layers by assuming plane waves is the simplest formalism in acoustics. We assume these plane waves such that they satisfy the motion equations, e.g. the ones introduced in eq. (1.1) or eq. (1.3). Introducing a simple problem, that of the scattering of an incident plate by a porous layer, we present an example of plane wave modelling for a simple system. The system is 2D with coordinates $\boldsymbol{x} = (x_1, x_2)$ constituted of a porous layer with thickness h and thus two interfaces of infinite extent along x_1, there is a fluid-porous interface at $x_1 = h$ and a porous-fluid interface at $x_1 = 0$. Both media of the air half-spaces and porous layer are modeled as fluid, thus the interfaces conditions should impose continuity of the pressure fields and normal particle displacements. The incident pressure field that satisfy the Helmholtz equation and originated from the lower interface takes the form of,

$$p_{0-} = a_+ e^{-ik_1 x_1} e^{-ik_{2,0} x_2} + R e^{-ik_1 x_1} e^{ik_{2,0} x_2}, \tag{1.13}$$

with a_+ the incident wave amplitude and R, the amplitude of the reflected wave. The pressure field in the porous layer is expressed as a sum of forward and backward waves,

$$p_p = b_+ e^{-ik_1 x_1} e^{-ik_{2,eq} x_2} + b_- e^{-ik_1 x_1} e^{ik_2 (x_2 - h)}, \tag{1.14}$$

and at the upper interface,

$$p_{0-} = T e^{-ik_1 x_1} e^{-ik_{2,0}(x_2 - h)}, \tag{1.15}$$

with T the transmitted wave amplitude. All wavenumbers should hold for the Snell-Descartes law that states

$$k_1^2 + k_{2,0}^2 = k_0^2, \text{ and } k_1^2 + k_{2,eq}^2 = k_{eq}^2. \tag{1.16}$$

The wavenumber should be imposed with $k_1 = k_0 \sin\theta$, $k_{2,0} = -k_0 \cos\theta$, with θ the angle of incidence, that governs the direction of propagation of the incident field. The displacement fields are additionally expressed using the momentum equation relating the latter with the pressure fields as $\partial_2 p = -\mathrm{i}\omega\rho_i u_2$, with $i = 0, eq$.

Next, the interface conditions describe the relationships between properties at the boundary between two media, including the continuity or vanishing of given fields. If a media is terminated by a surface (e.g. a rigid wall), a constraint should be applied on the fields to represent it, denoted as boundary conditions. Assuming a continuity between the two fluids in the present case with interfaces located at $x_2 = 0$ and $x_2 = h$, we obtain the following conditions,

$$p^t(h) = p^l(h), \quad u_2^t(h) = u_2^l(h) \quad p^l(0) = p^b(0) \quad u_2^l(0) = u_2^b(0), \tag{1.17}$$

with superscripts b and t denoting respectively the bottom and top interfaces.

The full analytical description of a system constituted from one or several interfaces can be done by writing a linear system built from the interface conditions, which vector of unknowns are the amplitudes of each of the existing waves in the system, that is $(b_+ \quad b_- \quad R \quad T)^{\mathsf{T}}$. The forcing right-hand side corresponds to the incident wave terms. The solution of the system yields all unknown amplitudes of the system, in particular the reflection and transmission coefficients. Imposing the interface conditions leads to writing the following matrix system,

$$\begin{pmatrix} 1 & e^{\mathrm{i}k_{2,eq}h} & 0 & -1 \\ \dfrac{k_{2,eq}}{\rho_{eq}} & -\dfrac{k_{2,eq}}{\rho_{eq}}e^{\mathrm{i}k_{2,eq}h} & 0 & \dfrac{k_{2,0}}{\rho_0} \\ e^{\mathrm{i}k_{2,eq}h} & 1 & -1 & 0 \\ \dfrac{k_{2,eq}}{\rho_{eq}}e^{\mathrm{i}k_{2,eq}h} & -\dfrac{k_{2,eq}}{\rho_{eq}} & -\dfrac{k_{2,0}}{\rho_0} & 0 \end{pmatrix} \begin{pmatrix} b_+ \\ b_- \\ R \\ T \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ a_+ \\ \dfrac{k_{2,eq}}{\rho_{eq}}a_+ \end{pmatrix}. \tag{1.18}$$

From this expression, the system can simply be solved to obtain all scattered amplitudes b_+, b_-, R and T, where R and T correspond respectively to the reflection and transmission coefficient. In the case of a rigidly-backed layer, the interface at $x_2 = h$ is replaced by a rigid termination. For (equivalent) fluids, a hard wall or Neumann-type boundary condition is translated by a vanishing normal particle displacement field along the boundary, that is $u_2(\Gamma) = 0$. The term depending on T thus vanishes, since p_{0+} does not exist; this reduces the size of the system.

This method is denoted in this work as the Global Method (GM), since all of the available data is used in the formulation of the problem as the unknowns are all wave amplitudes. As described in Section?? of Chapter ??, a general formalism can be used for this method and any stack of any layer can be described with the GM, by putting together a matrix system that contains all interface conditions in the system, in the general form,

$$\boldsymbol{M}(\omega)\boldsymbol{x} = \boldsymbol{f}, \tag{1.19}$$

where $\boldsymbol{M}$ corresponds to the interface conditions involved in the system, $\boldsymbol{x}$ contain the wave amplitudes and $\boldsymbol{f}$ is the forcing term, the incident plane wave amplitudes.

Similarly, a linear system can be written in the case of a PEM layer; the outer pressure fields remain the same but a solid and a fluid displacement field is modeled in the layer, added to the pressure field, as mentioned in Section 1.1. The interface conditions for a poroelastic-fluid coupling are thus,

$$\sigma_{12}^s = 0, \quad \sigma_{22}^s + \sigma_{22}^f = -p_0, \quad \sigma_{22}^f = -\phi p_0, \quad \phi u_2^f + (1-\phi)u_2^s = 0. \tag{1.20}$$

and for a rigid-backing,

$$u_1^s(\Gamma) = 0, \quad u_2^s(\Gamma) = 0, \quad u_2^f(\Gamma) = 0. \tag{1.21}$$

For the case of a rigidly-backed layer, the system is entirely written in in the appendices of Ref.[82]. For the symmetric layer, the linear system in given in Ref.[92]. Finally, the procedure to obtain the scattering coefficients is the same as previously.

A few results for the simple cases of the scattering of a plane wave by a rigidly-backed layer and in the case of a symmetric configuration (with air coupling at both interfaces) are represented by the diagrams in Figs 1.4a and b and results in Figs 1.4c and d, respectively. The GM results are presented by the continuous lines. There, the layer for both porous and PEM are both $h = 5$ cm thick and the incident wave propagates with a $\theta = 30^o$ angle. The parameters for each material corresponds to the one used in Table **??** from Chapter **??**.

The first case, also referenced to as a reflection problem compares the equivalent fluid and the full Biot model response. The two curves follow the same trend but additional effects due to the skeleton motion can be observed on the continuous lines while the equivalent fluid response shows to be smoother. For all configurations, the melamine layer is acoustically transparent at lower frequency and absorption can be noticed at slightly higher frequencies.

1.2.2. Transfer Matrix Method

The widely used method we know today as the Transfer Matrix Method (TMM), was initially developed by the works of Thomson[101] and Haskell[102] in order to describe elastic wave propagation in multilayer systems. This method can be applied to a range of wave physics among geophysics, quantum mechanics, electromagnetics and obviously acoustics[12,27,103] and elastic waves for ultrasonics[104]. Its principle lies in the evaluation of the state vector that regroups the variables of interest in the system and evaluate them at each interface, which is possible thanks to the interface conditions. Then, the physical field amplitudes stored in the state vector are propagated from one layer to the next in the case where multiple coupled layers are modeled. The transfer matrix $\boldsymbol{T}$ of a fluid layer, relating the pressure and normal displacement fields from the bottom (located at $x_2 = 0$) to the top interface (located at $x_2 = h$) of a fluid layer is thus,[12]

$$\begin{pmatrix} p(0) \\ u_2(0) \end{pmatrix} = \underbrace{\begin{pmatrix} \cos(k_{eq}h) & \mathrm{i}Z_{eq}\sin(k_{eq}h) \\ \mathrm{i}Z_{eq}^{-1}\sin(k_{eq}h) & \cos(k_{eq}h) \end{pmatrix}}_{\boldsymbol{T}} \begin{pmatrix} p(h) \\ u_2(h) \end{pmatrix}, \tag{1.22}$$

where k_{eq} is the bulk wavenumber of the wave propagating in the equivalent fluid and Z_{eq} is the medium impedances. The extension to PEM makes of a state vector that includes the fields, $\begin{pmatrix}\hat{\sigma}_{12} & u_2^s & u_2^t & \hat{\sigma}_{22} & p & u_1^s\end{pmatrix}^\mathsf{T}$. More details are given in Chapter 11 of Allard and Atalla textbook[11].

Stacks of N layers that have the same material nature are easy to manipulate, the transfer matrix of each layer has the same size as the others and thus,

$$\boldsymbol{T}_{\text{global}} = \prod_{n=0}^{N} \boldsymbol{T}_n. \tag{1.23}$$

TMM allows for an easy multiphysics coupling with interface matrices that relate the relations between the various fields at an interface which are thus non square matrices. This method was deep dived in several textbooks among which we propose Refs.[11,27] for additional details. In the last decades, there has been some works focused in the extension of this method to stabilize it and use for different applications and various formulations have been introduced like the Global Matrix Method**knopoff1964** which assembles the transfer matrices of all layers into a global equation system. The Stiffness Matrix Method has also been investigated and uses recursively applied compliance matrices to obtain a computationally stable formulation**rokhlin2002**. A different recursive approach has also been proposed by Dazel *et al.*[105] and introduces an Information Vector used to stabilize the method. Variations of the TMM also allows for the study of 2D system by combining transfer matrices for series elements and admittance matrices for parallel elements. This method has been used to model a Helmoltz Resonator (HR) embedded in an equivalent fluid porous layer[106].

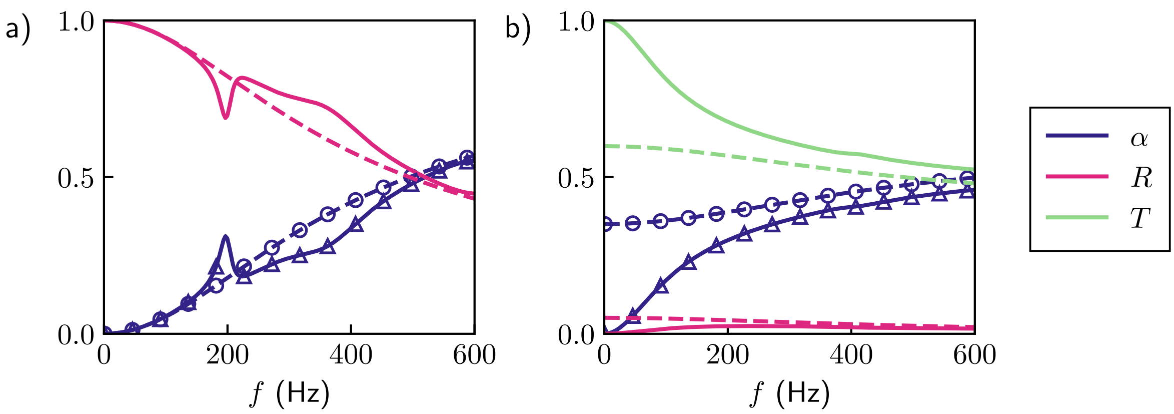

Fig. 1.4: Scattering coefficients from a poroelastic layer (continuous) and equivalent fluid layer (dashed) computed using the GM for the a) reflection, b) transmission problem. Results computed with the TMM are also shown for the absorption coefficient for the equivalent fluid (○) and for the poroelastic layer (△)

While very versatile, it remains unstable in limit cases at high frequencies or for large thickness of the layers. While the formulation is mathematically exact, the evaluation of the system involves involves finite-precision calculation which leads the very sensitive exponential terms to diverge in the previously specified regimes. From the evaluation of the transfer matrix of the global system, the scattering matrix can be obtained and thus the reflection and transmission coefficients by simply manipulating the expressions for the fields contained in the state vector with the scattering coefficients.

Considering a rigidly-backed layer, and thus $u_2(h) = 0$, we obtain a reflection coefficient,

$$R_b = \frac{T_{11} - Z_{eq}T_{21}}{T_{11} + Z_{eq}T_{21}}. \tag{1.24}$$

In the case of the symmetric layer, the reflection and transmission coefficients are determined from the scattering matrix as well,

$$T_s = \frac{e^{-ikL}}{T_{11} + T_{12}/Z_{eq} + Z_{eq}T_{21} + T_{22}}, \qquad R_s = \frac{T_{12}/Z_{eq} - Z_{eq}T_{21}}{T_{11} + T_{12}/Z_{eq} + Z_{eq}T_{21} + T_{22}}. \tag{1.25}$$

which is only valid for reciprocal and linear assumptions. The absorption coefficient for the same two configurations are calculated as $\alpha_b = 1 - |R_b^2|$ and $\alpha_s = 1 - |R_s^2| - |T_s^2|$ and are shown in Fig. 1.4with markers. It can be seen that the same results (up to the respective numerical precision of each method) are obtained. No absorption peak is observed in the results since the frequency range is way below the quarter-wavelength resonance (which should lie around 1.5

kHz for these 5 cm-thick homogeneous layers). The equivalent fluid and the full Biot theory results follow generally the same trend for the results of the rigidly-backed layer in Fig. 1.4a, except for a peak around 200 Hz shown with the poroelastic model. This is known as the Biot peak and corresponds to an elastic resonance, a strong vibration of the skeleton leading to increased structural dissipations, hence the reflection coefficient dip.[82,107]. The results in the transmission case in Fig. 1.4b present a generally lower absorption, bounded at $\alpha = 0.5$. Here, the equivalent fluid and poroelastic respective response are significantly different: at low frequencies, the Biot theory behaves as expected with the transmission close to 1 and the reflection close to 0. The equivalent fluid reflection coefficient is similar but its transmission lies around 0.6 for lower frequencies, leading to a rather important absorption coefficient at zero frequency. While not representing a physical and observable behaviour of the system, this effect is a feature of the JCA model due to the low-frequency limit of the real part of the effective density. A correction term has been proposed by Pride *et al.*[108] to improve this low-frequency limit[11].

1.3 Dispersion of guided waves in layers

In this section, we take an interest in similar configurations as previously, removing the excitation or forcing term in the equations. Solving the appropriate set of equations for a given set of parameters leads to the modes of the free system, that is the solution of the characteristic equation leading a specific behaviour of the wave propagation in the structure for the said parameters. The form of the modes of wave propagation in layer is discussed and various methods for their calculation of are presented. The existence of an interface in the geometry leads to the propagation of the so-called surface waves[109] where the interface itself acts as the support for propagation.

1.3.1. Example of a planar air waveguide

A dispersion relation can be illustrated using the example of an elementary planar waveguide. A fluid layer of infinite extent in one direction, with thickness h, and Neumann-type boundary conditions at each end $x_2 = 0$ and $x_2 = h$. This configuration is the simplest type of waveguide where waves are guided by successive reflection onto each wall. This leads to modes of the system in the form $k_{2,n} = n\pi/h, \forall n \in \mathbb{N}$[12,110]. Finally, the dispersion relation writes as,

$$k_{1,n}(\omega) = \sqrt{\left(\frac{\omega}{c_0}\right)^2 - \left(\frac{n\pi}{h}\right)^2}, \tag{1.26}$$

with c_0 the velocity in air. The solutions to this equation for a range of $n = 0$ to $n = 4$ is shown in Fig. 1.5. There, the wavenumber appears as either purely real or purely imaginary, indicating respectively the propagative and evanescent modes of this simple geometry. The mode with order 0 is the one that exists in unbounded fluid materials, with its slope corresponding to the fluid velocity c_0. The next orders all present a cut-off frequency below which the modes become evanescent.

Different types of waves propagate along the interface. An example can be water waves that follow the interface between air and water; most of the energy is localized at the interface. A

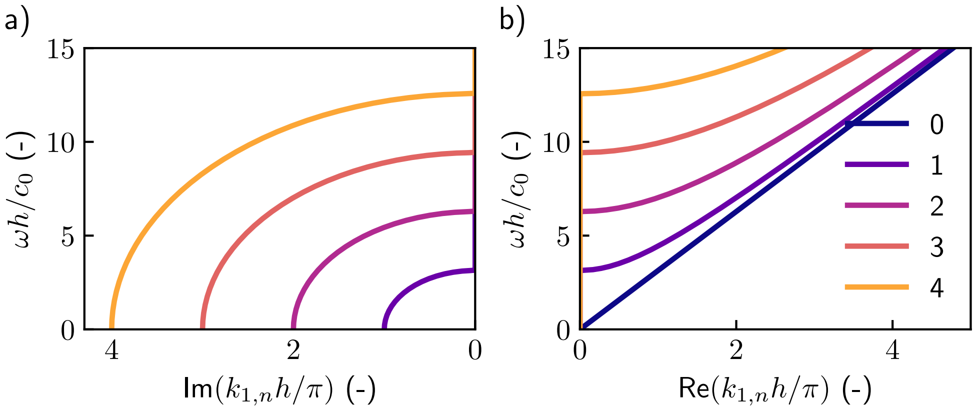

Fig. 1.5: Adimensional dispersion relation of an air waveguide with thickness h = 5 cm. a) Real part of the wavenumber and b) Imaginary part of the wavenumber

similar behaviour exists for seismic waves where the interface of a solid substrate supports the propagation of Rayleigh (longitudinal motion) and Love waves (transverse motion) which both show an energy strongly decaying with depth. Scholte waves are waves that propagate along the interface between a fluid and a solid medium. In fluid layers, the propagating waves are similar to studying a shear horizontal (SH) wave in an elastic layer. One mode propagates at zero frequency, while countless other modes propagate with a cut-off frequency that corresponds to the successive quarter-wavelength resonances, essentially depending on the layer thickness and the nature of the boundary/interface conditions.

From the previously obtained linear systems in Section 1.2.1 for the configurations, and removing the incident wave from the system, i.e. the forcing (right-hand side) term, an homogeneous linear system is obtained, in the general form $\mathcal{L}(\boldsymbol{k}, \omega)\boldsymbol{s} = 0$, where the operator $\mathcal{L}$ governs wave propagation in the structure, $\boldsymbol{k} = (k_1, k_2)^\top$ is the wavenumber vector in the 2D geometry and $\boldsymbol{s}$ a vector containing the amplitude of all the partial waves. In order to obtain nontrivial solutions for $\boldsymbol{s}$, the system should satisfy a vanishing determinant for the linear system, $\det(\boldsymbol{L}(\boldsymbol{k}, \omega)) = 0$. The parameters satisfying this so-called *dispersion equation* can be sought, allowing to identify the modes of the structure. These equations can be solved for various parameter spaces depending on the perpective adopted on the problem[111,112]: solving for frequencies while imposing the wavenumber or the other way around. The first one is useful when interested in the time-dependant effects the structure has on the propagation. In this work, we are interested in the losses generated by the structure and the material and as such frequency is fixed in order to solve complex wavenumbers. The solutions are thus able to indicate on rather the spatially induced dispersion occuring in the system.

1.3.2. A bestiary of wave solutions in dissipative and open waveguides

In this section, an overview of the possible sets of solutions for the guided waves in waveguide is presented. These sets of solutions are determined by the direction of the propagation, the attenuation and the nature of the leaky wave if it exists. This *bestiary* is split into 3 categories. Considering a layer and half-space system, there exists three wavenumber components from Eq. (1.16). The form of solutions are assumed as

$$p_l \propto e^{-ik_1x_1}e^{\pm ik_{2,eq}x_2}, \quad p_0 \propto e^{-ik_1x_1}e^{\pm ik_{2,0}x_2}, \tag{1.27}$$

with $k_1 = +|k_1'| - i|k_1''|$ and $k_{20} = \pm|k_{20}'| \pm i|k_{20}''|$. The longitudinal wavenumber is the same in both layer and half-space domain as a consequence of the Snell-Descartes law and the continuity equations: there exists in total three dependant wavenumber components in this system, k_1, $k_{2,0}$ and $k_{2,eq}$ among which determining two allows to compute all,

$$k_{2,0} = \pm\sqrt{k_0^2 - k_1^2}, \quad k_{2,eq} = \pm\sqrt{k_{eq}^2 - k_1^2}. \tag{1.28}$$

An important relation between these wavenumbers can be obtained by taking the imaginary part of Eq. 1.16,

$$k_1' k_1'' = -k_{20}' k_{20}''. \tag{1.29}$$

A set of physically meaningful types of solutions should satisfy this equation,[113]; the following section will provide a detailed discussion of this issue.

Because of the symmetrical nature of the presented configurations, and the considerations of unbounded layers in the x_1 direction, a plane of solutions corresponding to $k_1' \geq 0$ can be selected, as the right-going solution is perflectly equivalent solutions to the left-going solutions.

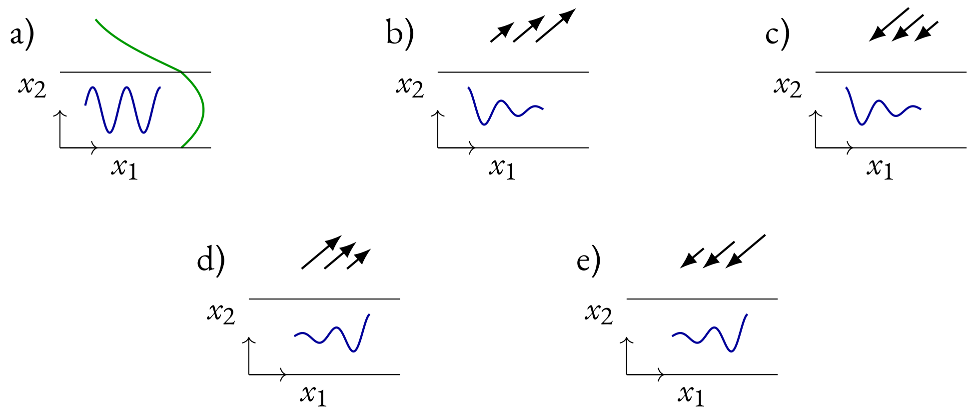

Fig. 1.6: Classes of solutions for the different types of propagating waves in open non-dissipative waveguides a) perfectly guided solution with the amplitude of the field along x_2 displayed in green: the solution for the field in the surrounding medium is purely evanescent b) leaky growing waves with an attenuated solution in the layer and growing outgoing solution in the surrounding domain, c) backward leaky attenuated waves corresponding to an ingoing solution, d) outgoing growing wave solution (improper) and e) incoming growing waves (improper).

In **conservative and closed systems**, the study is restricted to only the two wavenumber components in the layer, since no half-space is accounted. The total acoustic energy is thus confined to the structure enclosed by its boundaries. A type of wave solution exists as *perfectly guided*, this propagating wave has $k_1' > 0$ with no attenuation $k_1'' = 0$. Solutions include *purely evanescent* wave as well $k_1'' > 0$ which often exists and continuates a propagating mode below its cut-off frequency as in simple waveguides configurations as presented in Section 1.3.

Open systems support the existence of the previous types of modes with additional phenomena. Since part of the energy is susceptible to radiate out from the structure, the energy is no longer confined to the layer. Some of the previously purely guided modes can thus become attenuated, $k_1'' < 0$. Waves are still *perfectly guided* with conditions $k_{20}' = 0$ with evanescent waves in the fluid $k_{20}'' < 0$, meaning that waves are confined to the proximity of the layer interface as shown in Fig. 1.6 with the amplitude along the x_1 direction in blue and the amplitude along the x_2 direction. The additional condition $k_1' = 0$ correspond to *trapped modes*, which are a set of discrete eigenvalues of the system. They can exist in waveguides where the system is confined in one direction at least, the energy can thus be localized around some obstacle or defect by the evanescent waves[114,115]. In open systems, they correspond to waves that are evanescent at infinity. However, in open waveguides, the acoustic energy is usually able to radiate outside of the waveguide, leading to two type of waves as *leaky waves* in the case $k_{20}' > 0$ and *incoming waves* (also called backward leaky) with $k_{20}' < 0$. These solutions are respectively exponentially growing as it radiates $k_{20}'' > 0$[116,117] (see the leaky growing wave represented in Fig. 1.6b and incoming growing wave in Fig. 1.6e) or attenuated $k_{20}'' < 0$ (see the leaky attenuated wave represented in Fig. 1.6c and incoming attenuated wave in Fig. 1.6d). These at first counterintuitive results ensue from the requirements that the wave should be propagating $k_1' > 0$ and attenuated $k_1'' < 0$ along the layer, and satisfying Eq. **??**. The book of Collin[85] can be refered to for fundamental discussions about the location of these solutions in the complex plane. We also mention Section 3.2 from Kiefer's thesis[118] which is dedicated to the form of solutions for quasi-guided waves in the case of an elastic layer coupled to a semi-infinite fluid half-space.

The addition of **dissipation in open systems** leads to the mixing of several loss mechanisms: the leakage of waves and the dissipations within the guide (which can be of whatever nature). In this case, the separation between incoming and leaky waves is not possible anymore. This particular aspect has been very scarcely tackled in the literature: an unpublished article is interested in an elastic waveguide[93] while in optics, a paper discusses the waves guided in nanospheres with a complex permittivity[119]. There, the modes are sorted as *proper* and *improper* solutions. This complex case is covered in a dedicated section **??** in chapter **??**.

In the next parts of this chapter, a discussion on available methods to compute the complex dispersion relation for guided waves in single or stacks of homogeneous layers is proposed. There are semi-analytical methods that can be used, Section 1.3.3 or numerical methods that involve some sort of discretization in Section 1.3.4.

1.3.3. Modes of porous layers computed with semi-analytical tools

The literature on the dispersion relation of porous/poroelastic-based structures is scarce; Groby *et al.*[120] have shown the complex dispersion relation in a homogeneous equivalent fluid layer coupled to air while Boeckx *et al.*[92] have computed the modes propagating in a poroelastic waveguide coupled at each top and bottom interface with an outgoing plane wave propagating in an air medium. This has been done using an analytical description of the system by expressing the interface conditions in a linear system (which we call GM in this work). Both of these work have used a root-finding procedure in order to solve the modes of the structure.

Using Eq. (1.19), the non-trivial solution of the problem requires that $\det(\boldsymbol{M}(k_1, \omega)) = 0$. The determinant of the matrix from Eq. (1.18) can be further expressed as,

$$2\frac{k_{2,eq}k_{2,0}}{\rho_{eq}\rho_0}\cos(k_{2,eq}h) + \mathrm{i}\left(\frac{k_{2,eq}^2}{\rho_{eq}^2} + \frac{k_{2,0}^2}{\rho_0^2}\right)\sin(k_{2,eq}h) = 0. \tag{1.30}$$

This expression can not be solved analytically. Solving wavenumbers can be done through the writing and solving of a (possibly generalized) eigenvalue problem or by root-finding of solutions in the case where writing the first one appears impossible. From this expression, one can use a root-finding procedure to solve for either the term $k_{2,eq}$ or k_1 to obtain the dispersion relation by varying the frequency. Results are shown in fig. 1.7.

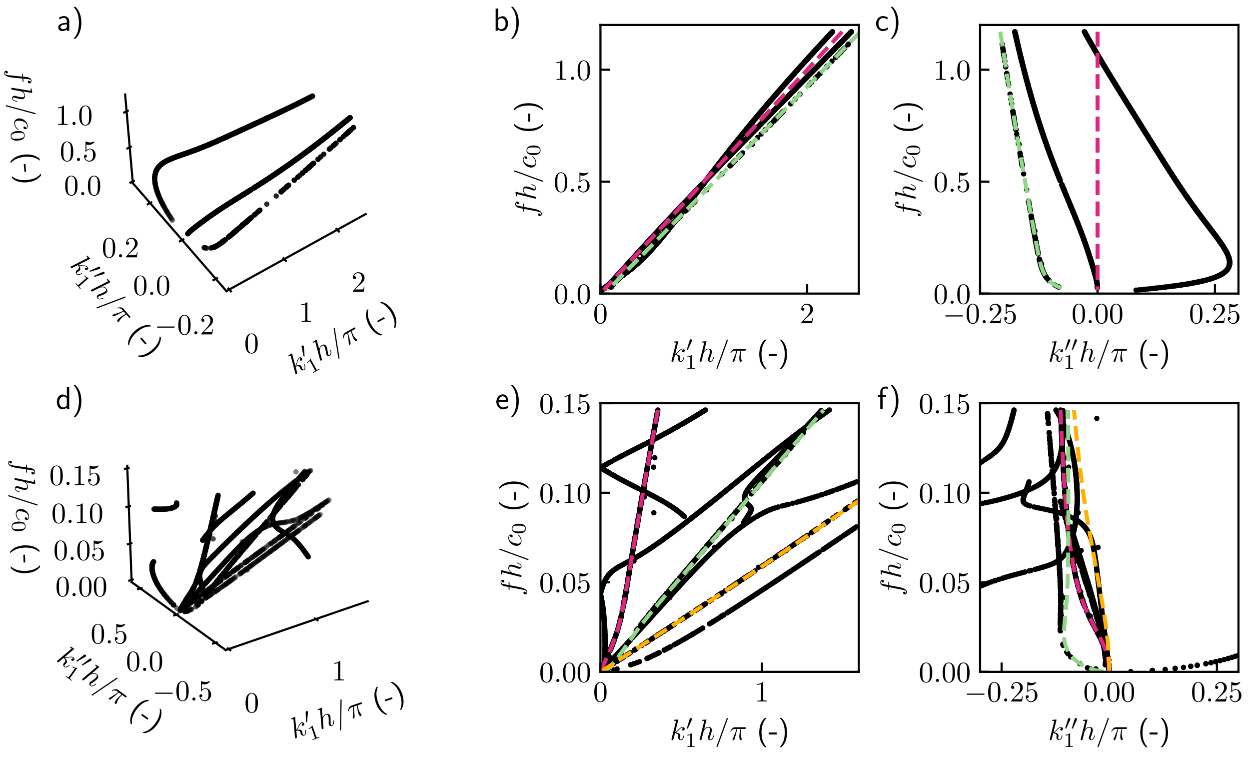

Fig. 1.7: Dispersion relation of a porous layer loaded by an air half-space on both sides. a) 3D view with k_0 (– –) and k_{eq} (– –), b) k_1' plane , c) k_1'' plane. d-f) Same views in the case of the poroelastic layer with k_p (– –), k_a (– –) and k_s (– –).

We reference to Appendix ?? for extensive details on the root-finding procedure based on the Müller method. These results show an infinite amount of modes that have increasing attenuation, only the first four are shown in Fig. 1.7a to Fig. 1.7c. A branch corresponding to k_{eq} is found by the root-finding procedure, as setting $k_1 = k_{eq}$ in Eq. 1.30 indeed leads to a trivial solution. Only one propagating mode exists in the frequency range shown (one branch lies between the lines k_0 and k_{eq}) as is the expected behaviour of a lossy fluid layer.

A similar configuration, with a layer modeled using the full Biot theory is shown in Fig. 1.7d to Fig. 1.7f. Results are significantly different as the dispersion relation presents a large amount of branches. Alike the equivalent fluid results each bulk wavenumber constitutes a trivial solution of the characteristic equation. Since the elastic motion of the layer is modeled, quasi-Lamb modes are obtained in this case, which are all complex because of the viscoelastic and viscothermal dissipations in the poroelastic layer. In this case, the modes are not simply separated into evanescent and propagating modes. When their attenuation is low, they are quasi-guided: corresponding to propagating solution with some attenuation. These results are comparable to those of Boeckx *et al.*[92] and to those presented in Section ??.

Generally, characteristic equations as Eq. 1.30 can only be obtained for very simple cases as is the case here, for a single fluid layer. The characteristic equation for Lamb modes is also well known[110]. However, more complicated geometry requires to solve the root-finding iterative process by using directly a numerically-computed matrix determinant which is computationally very intensive, or in many case, it is outright not possible due to the analytical complexity of the description of the system. One of the main downside of these methods is that they require multiple initial guesses to construct an interpolating function.There is no assurance of finding a solution in a given location: depending on the solution space smoothness and shape, the iterative process might get stuck in local minima, or can simply not converge in some situations. In cases where a complete dispersion relation is complicated to be constructed, more complex numerical method could thus be required; a discretization of the geometry and an interpolation of the physical fields is done as will be discussed in the upcoming section.

1.3.4. Numerics for modes computation

Several numerical methods have been investigated in the last decades with the goal of computing complex modes in homogeneous layers. Guided waves are of interest in a wide range of geometries and we focus here on the case of unbounded single (or stack of) layers with two infinite interfaces as tackled in the previous sections.

Even if a semi-analytical description of the system is possible, it can be advantageous to be able to write the problem differently, as root-finding routines often lead to challenges as discussed in earlier sections. With that in mind, several numerical methods have been introduced with the same framework: a 1D discretization of the layer along the thickness and assuming harmonic motion along the longitudinal direction.

Modeling of the semi-infinite radiating space

There is a discussion to be made on the relevance of the strategy to use for the radiating medium modelling. In the literature, two have been presented: either the coupling can be imposed locally at the interface condition, or the domain of the surrounding medium is fully modeled, accounting for some termination condition satisfying a Sommerfeld radiation condition, by using a Perfectly Matched Layer (PML) or some analogous absorbing layer. The use of artificial absorbing layers and the discretization at the end leads to the apparition of a continuum of modes of the cavities[121] which requires to be sorted out. A strategy based on a local coupling between the structure and the radiating medium leads to impose a form of solution for the radiation, usually as plane waves[122]. An alternative for local coupling has been investigated by associating a Boundary Elements Method (BEM) formulation for the interface[123].

We argue that the advantages of the imposition of a local coupling is two-fold: it requires less degrees of freedom for any sort of discretization since the interface condition describes the full motion of the radiated waves. For dispersion problems, this is particularly interesting since leaky mode solutions are thus directly obtained, without needing to filter out all radiation modes linked to the additional cavity and the resonances of the PML. Also, as explained in Section ??, leaky waves are characteristic solutions of open waveguides that show exponential growth as they radiate away from the layer, which is especially ill-suited for implementing PMLs for this kind of system, although it has been achieved[121,124]. We again refer to the thesis of Kiefer[118], especially section 3.1, for a in-depth discussion of modeling the full infinite medium against only an interface condition coupling with the layer.

Numerical methods used in the literature

Spectral methods have been around for quite some time and have been used for many numerical applications. We focus for this work on Spectral Collocation Method (SCM)[125], which has been especially used for the computation of guided wave modes in elastic layers. One of the first paper that introduced this method for the computation of complex dispersion relations is that of Adamou *et al.*[126], while other have focused on multilayer anisotropic structures[127] or cylinders[128]. These references propose an implementation for closed systems only. We refer to the introduction of Chapter **??** for an extensive bibliography on SCM applications.

This method is useful for the discretization of 1D geometries as is the object of the present section. In the case of simple 2D geometries, a 1D discretization along the transverse direction associated to a harmonic motion in the longitudinal direction. Even if the use of finite elements seems farfetched, this method could be relevant in contexts where boundary effects or guide with varying cross-sections are studied. However, a variation of finite elements based on a 1D discretization of the waveguide is entitled Semi-analytical Finite Elements (SAFE) and is akin to SCM which we have described in the previous paragraph. The 1D discretization is done along the thickness of the layer by regular nodes, and the fields are interpolated using finite elements shape functions. Examples of Semi-analytical Finite Elements (SAFE) use cases are very similar to SCM and include. A detailed description of the calculation of Lamb modes using a simple SAFE implementation using high-order Lobatto polynomials is proposed in Appendix. **??**. There, its performance is also compared to that of SCM. The particular advantage of SCM is the very quick convergence allowed by the interpolation using Chebyshev polynomials, called *spectral convergence* in the literature[125,129,130]. As it was shown in the aforementioned appendix, we saw that it outperforms higher-order SAFE. Thus, we will focus on this method for the modeling of multilayer homogeneous structures.

Open elastic layers accounting for surrounding air media that couples the elastic layer have been first presented by Kiefer *et al.*[122] where a SCM was used to discretize the elastic layer. Additionally, a local coupling is modeled with a discretized interface conditions. The system of equations is then formulated as a GEP and can be solved with traditional eigenvalue solvers. This work forms the basis of the work done in Chapter **??**. Other implementations of the same configuration include a discretization of the fluid half-space by means of a mapping allowing the numerical solution to increase exponentially at infinity, as is a characteristic of leaky waves[131].

With Semi-analytical Finite Elements (SAFE), which has been been widely used for similar configurations as well have computed modes of immersed waveguides[124], leaky Lamb modes using PMLs and a coupling between SAFE and BEM[123]. A detailed overview on the literature dealing with this topic is proposed in Appendix **??**. Ducasse and Deschamps[111] have also formulated an eigenvalue problem by discretizing the transverse direction of the layer using finite differences.

1.4 Methods for modelling metaporous surfaces and interfaces

1.4.1. A word on periodic structures

A large literature that shows the enhancement of absorption properties by periodicizing porous layers with scatterers was discussed in the introduction. In this section, a presentation of the fundamental notions surrounding periodic structures is proposed. In wave physics, periodic structures have been shown to introduce interesting effects towards applications for wave control and constitute the cornerstone of acoustic metamaterials. The periodicity can be imposed in acoustic systems using the so-called Bloch-Floquet theorem. This formalism has been pioneered by Felix Bloch and Léon Brillouin among others[86,132]. Throughout this manuscript, we consider a periodic cell along only the longitudinal direction, that 1D periodicity which we denote d. Taking an infinitely repeating lattice or a crystal, the description of the wave propagation in only one of the so-called unit cell, one spatial period of the total structure, is enough to describe the whole structure. The layer of infinite extent is thus limited to a single unit-cell. A wavefunction $\psi(\boldsymbol{x})$ describing the propagation should thus depend on a field with the same spatial periodicity as the lattice. To do so, the wavefunction is projected onto an infinite modal basis, that is Bloch modes,

$$\psi_q(\boldsymbol{x}) = e^{\mathrm{i}k_q x_1}. \tag{1.31}$$

The wavefunction at the left boundary is thus related to that at the right boundary modulated by the spatial periodicity and some translation vector, that is $\psi(x + qd) = e^{\mathrm{i}\boldsymbol{k}}\psi(x)$.

The possibility of reduction of the spatial considerations of the problem also leads to a reduction of the reciprocal wavenumber space. In periodic structure, we often restrict this reciprocal space to the first Brillouin zone, that is to wavenumbers restricted to $k_{1q} \in [0, \pi/d]$ when looking at the band structure. The consequences of periodic media are directly observable in the band structure: bands are usually folded when the wavenumber reaches this limit of the reciprocal space. This folding occurs at the Bragg frequency corresponds to a relation between the wavelength λ of the excitation signal and the periodicity of the crystal lattices, $d\sin(\theta) = q\lambda$, with θ the angle of incidence. This condition corresponds to destructive interferences: this phenomenon is a cause of the appearance of the famous Bragg bandgaps; in these bands of frequency, waves are unable to propagate[25,133]. The study of the combination of damping and dispersion in 2D phononic crystals was studied in Ref.[134]. Variations of these phenomenon have been investigated in acoustic metamaterials: e.g. the addition of periodic resonators leads to hybridization stopbands[**krasikova2024**, 135].

1.4.2. Scattering of waves by periodic structures

Multiple scattering

Analytical methods for the evaluation of the wave propagation in periodic structures often require tedious derivations, as the frequency regimes of interest is usually high enough to require the description of the multiple scattering happening in the structure. Simply explained, the scatterer in the unit cell onto which the incident wave impinges acts as a secondary source, which scattered field is successively scattered by each neighboring obstacles and so on. This theoretical framework is called the Multiple Scattering Theory has been developed to account for these phenomena in various disciplines.[27,136]. Considering an array of periodic cylindrical scatterers onto which a plane wave impinges, the form of the pressure field is,

$$p(\boldsymbol{r}) = \sum_q A_q G_q + \underbrace{\sum_j \sum_q B_q^{(j)} G_{jq}(\boldsymbol{r}_j)}_{\text{scattered field}}, \tag{1.32}$$

with q the Bloch mode index, G_q is a function involving Bessel function and depends on the angle of incidence of the plane wave. The scattered field represents the sum of the contributions of the scattered field by each jth cylinder with $G_{jq}(\boldsymbol{r}_j)$ being a function that depends on first-kind Hankel functions, defined in the local cylindrical coordinate system of the jth scatterer. The unknowns of the system are the $B_q^{(j)}$ amplitudes.

This method has been used to compute the acoustic response of metaporous structures with rigid inclusions[71] as well as metaporoelastic surfaces with solid and shell inclusions[82]. Both configurations have been tuned to show a perfect absorption peak in the frequency spectrum. The challenge of the method lies in the formulated of the expressions, but leads to a system of equations that is computed at a much lower computational cost than most numerical methods, e.g. compared to finite elements.

Finite elements

Most of the literature covering the scattering of waves by periodic scatterers and specifically metaporous structures use FEM, for some part through the interface of commercial software like Comsol Multiphysics or Ansys. This very versatile and powerful method has been used for decades in all fields of physics and thus won't be detailed in too much details this thesis. It is recommended to the reader as an introduction on the basis of FEM, the textbooks from Atalla and Sgard[137] related to acoustics and vibration problems and that of Dhatt *et al.*[138] and Solin *et al.*[139] dealing specifically with higher-order finite elements problems. This method allows for the discretization of the geometry via a mesh in order to solve the appropriate motion equations. This has allowed to use rather complex geometries, e.g spirals[140] and labyrinthine paths[68] as scatterers embedded in porous layers as discussed in the state of the art. Additionally, Gaborit *et al.*[83] have developed a method in order to couple FEM with Bloch modes expansion of analytical fields described by TMM which also been done in Ref.[141]. They have used it to compute the scattering coefficients of structure that include a periodic cell sandwiched by homogeneous layers with an embedded inclusion.

Other variations of the FEM include the so-called p-FEM or higher-order finite elements, that uses polynomial Lobatto shape functions instead of the traditional Lagrange shape functions. Several works have highlighted it as being a very powerful discretization procedure[142,143]. The Wave Finit Elements (WFE)**<empty citation>** + dgm, bem and

In this section, we propose a summary of the Finite Elements Method (FEM), presented in the case of fluid domains. A simple derivation is done to describe generally the procedures involved in the method. Let us start with a simple Helmholtz equation governing the acoustic motion in a general domain Ω,

$$\nabla^2 p + k_{eq}^2 p = 0 \quad \text{in } \Omega. \tag{1.33}$$

This equation is associated to the appropriate boundary conditions prescribed on the contour of the domain $\Gamma = \partial\Omega$. Using the Galerkin method, this strong form is cast into a variational (also called weak) formulation, as

$$\int_\Omega k_{eq}^2 \delta p \cdot p - \boldsymbol{\nabla}\delta p \cdot \boldsymbol{\nabla} p \,\mathrm{d}\Omega + \int_{\partial\Omega} \delta p \partial_n p \,\mathrm{d}\Gamma = 0, \quad \forall\, \delta p \tag{1.34}$$

where δp is an admissible set of test functions associated to the pressure field. primal and dual here

The discretization in FEM always requires the computational domain to be split into a set of small non-overlapping elements (in the present case, triangular). Within each element e with domain Ω_e, the pressure field is approximated by a finite sum of local shape functions N_n^e and associated nodal degrees of freedom p_n^e.

In this work, we use Lobatto shape functions. For 2D elements, there exists nodal, edge and bubble shape functions[143,144]. At the nodes, interpolation on the interval $\xi \in [-1, 1]$ is done by using linear (Lagrange) nodal shape functions. For higher-order approximations, that is with $p > 1$, these shape functions are enriched with the so-called bubble shape functions constructed from integrated Legendre polynomials which are internal shape functions, called Lobatto shape functions. This construction leads to a hierarchical basis that spans internal nodes within each element.[142]. The pressure field in a single element can thus be discretized as,

$$p_e(\boldsymbol{x}) \approx \sum_{n=1}^{n_{dof}} N_{n,e}(\boldsymbol{x}) p_{n,e} = \boldsymbol{N}_e(\boldsymbol{x})^\top \boldsymbol{p}_e, \tag{1.35}$$

where $\boldsymbol{N}_e$ is the matrix containing the shape functions and $\boldsymbol{u}_e$ is the vector containing the nodal values. For each element, the local integral formulation is mapped from the physical domain (x, y) to the reference element domain (ξ, η). Thus, the local solution of an element can be computed from the reference frame and mapped into the physical space.

In a single element, the discretized volume term is expressed as,

$$k_{eq}^2 \int_{\Omega_e} \delta \boldsymbol{p} \boldsymbol{N}^\top \boldsymbol{N} \boldsymbol{p} d\Omega_e - \int_{\Omega_e} \delta \boldsymbol{p} \nabla \boldsymbol{N}^\top \cdot \nabla \boldsymbol{N} \boldsymbol{p} d\Omega_e = 0, \tag{1.36}$$

which can be expressed as a matrix system with each matrix having size $2p + 1$. In summary, the finite element solution procedure starts from the weak formulation of the governing PDEs. The domain is discretized using triangular elements, shape functions are defined on a reference element (with Lobatto basis functions used for higher-order interpolation), and the global system is constructed via numerical integration and assembly. This methodology is the foundation for deriving the discrete system equations (e.g., Eq. (22)) discussed in Chapter 4.

This formulation can be systematically extended to other physical models. For instance, applying the same Galerkin procedure to the linear elasticity equations yields the weak form used for solid domains (see Eq. ?? in Appendix ??).

1.5 Dispersion relations and band structure in periodic lattices

1.5.1. Methods for non-radiative systems

One of the most popular methods for the computation of band structures in periodic systems made of classical lossless materials are the Plane Wave Expansion (PWE) which is semi-analytical. It relies on a Fourier transform of the motion equations, which allows an expression of the fields expanded onto Bloch modes, truncated for numerical applications. This method, while easy to implement presents important limitations due to the convergence of Fourier series and the matching of constituent materials for the lattice: materials should preferably of the same nature for the calculation to go well[27]. Generalized as the Extended Plane Wave Expansion (EPWE), the method also allows to obtain the evanescent Bloch modes in Ref.[145], allowing for complex dispersion diagram investigation and to account for evanescent modes in finite 2D periodic lattices.

The complexity of the system prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods. As the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation. The thesis of Magliacano[146] was focused in the modeling of a closed system, of an equivalent fluid unit cell with an embedded rigid inclusion. The finite elements formalism was entitled with the so-called shift cell technique, that implements the Bloch-Floquet phase shift at the boundaries inside of the constitutive variational weak formulation.

1.5.2. Unpromising adaptation of existing methods

The complexity of the system of interest in this thesis as presented in Section?? of the Introduction prohibits the use of such semi-analytical methods. For more complex geometries, we must turn to alternative methods: as the 1D discretization used by the SCM and SAFE and the plane wave formalism used in TMM and the GM is not capable of providing a description of the geometry of inhomogeneous layers, in our case, the effects of the scatterer in the unit cell on wave propagation.

Multiple scattering could also be considered. However, it has been successfully used to describe the acoustic scattering of periodically arranged rigid inclusions in equivalent fluid unit cells[71] as well as solid inclusions in a poroelastic unit cell[82]. However, the so-called Schlömilch series that appear in the calculation, from the effect of the scattering of the neighboring cylindrical inclusions appear to be not so manipulable when considering complex longitudinal wavenumbers. Its expression is

$$\sigma_q^X = \sum_{l=1}^{\infty} H_q^{(1)}(k^X ld)\left((-1)^q e^{ik_1^X ld} + e^{-ik_1^X ld}\right), \tag{1.37}$$

and indeed, separating the k_1^X real and imaginary inevitably leads to at least one growing exponential term in the series, leading to it being unable to converge, up to our study.

We face challenges when trying to model this structure using the tools classically used in the literature, that is COMSOL Multiphysics. Although there are built-in tools in the software for "*Eigenfrequency*" studies, the library aims at solving complex frequencies of the system, given a set of parameters for the wavenumber. As it has been discussed earlier in the chapter, the goal of this work is to solve for complex wavenumbers and real frequencies. Even trying a manual implementation of the model in the "*Weak Form PDE*" interface, the tendency of this software to function as a black box introduces a new degree of complexity to this strategy. While FEM remains the soundest strategy for the modeling of the complex inhomogeneous geometry of the unit cell of metaporous surface, we propose to adapt in-house existing frameworks in order to develop numerical tools in this thesis.

1.6 Conclusion and prospects of methods for the thesis

In the case of homogeneous multilayer systems with surrounding fluid loading, we have seen that semi-analytical have long been established in the literature in order to compute their scattering properties. For dispersion problems however, these configurations usually require some sort of numerical discretization to treat the geometry; a wide range of methods allowing for the direct formulation of an eigenvalue problem and a computation of modes was shown in Section 1.3 with first semi-analytical tools applicable to simple cases (e.g. an equivalent fluid layer with surrounding air), and then various numerical methods like SCM or SAFE as the two main protagonist.

When going to periodic systems, it appeared that the way to tackle the problem is quite different. Indeed, the resulting complex geometry (due to the obstacles) leads to much different tools. In the case of scattering properties computation, semi-analytical tools are still available. We have seen that FEM seems to provide promising possibilities for the modelling of complex periodic cores. However, the considerations of open systems leads to question regarding the modeling of the half-spaces surrounding the structure. Two strategies compete in the literature: either a fluid cavity is added to the geometry and terminated by some sort of absorbing layer at the end; or the harmonic fluid radiation is locally accounted for as an analytical interface condition of the structure. There is also an interest in the coupling of poroelastic homogeneous layers or periodic configurations with additional layers to model plates or sandwich structures and coatings.

Conclusion & Perspectives

This thesis work has proposed the development and validation of several numerical tools allowing to model metaporous surfaces and interfaces. These structures feature dissipation mechanisms, strong dispersion due to the periodicity as well as leakage, thus involving challenges in the modeling process. Methods were developed to compute the dispersion relations of leaky guided waves propagating in homogeneous multilayer structures, evaluate the scattering properties of grating stacks and compute dispersion relations of waves in metaporous layers.

Throughout the literature, these metaporous structures have shown efficient absorption properties using various geometry by embedding resonant, tortuous paths, voids or inclusions inside porous and poroelastic layers. These obstacles provide additional dispersion as well as interferences of the wave at discrete frequencies. Combined with the attenuation of waves in porous layer, they provide for efficient acoustic absorption structures. Even though, these effects have been studied in the literature, they have mostly been considered as infinite and closed systems. In this work, the considerations were extended to the study of leaky guided waves radiating to surrounding media. The combination of these loss mechanisms thus have showned a complex behaviour featured by the structure and that the classification of the waves was of interest and that the physical mechanisms at stake were not so simple.

The foundation for the work was laid out in CHAPTER 1, introducing the governing motion equations for the assumptions of homogeneous porous and poroelastic media and proposed a state of art for the exisitng numerical method allowing for the computation of the scattering properties and the dispersion of waves in homogeneous layers and periodic systems.

Next, CHAPTER 2 deals with dissipative homogeneous per layer multilayer structures accounting for leakage. These homogeneous layers can be described by prescribing a harmonic motion in the infinite direction while using the SCM and a 1D discretization to describe the propagation along the transverse direction. With this general formalism, any stack of layer, dissipative or not, can be computed with the presented method. Results regarding the energy flux for specific modes were presented as well. This method was implemented and validated as a fast and accurate technique for reliably computing the dispersion relations. Its efficiency and stability were demonstrated through comparisons with semi-analytical results. Results include the study of a two-layer structurefinir ça.

In CHAPTER 3, a robust and stable method for dealing with grating stacks constituted of poroelastic and porous layers. finite element discretization remained stable for all tested cases contrary to the TMM and the coupling is thus quite robust and efficient, since no fem discretization of the homogeneous layers is required.

In the final CHAPTER 4 we adapted the previous method, in the absence of a source, solving its dispersion relation. The method relies on the same discretization as previously, therefore the root-finding solving of the method is not very efficient. However, reliable results have still been achieved. We were able to highlight the trapped mode that occurs in a perfect absorption case as well as the impact of the resonance due to an hybrization of the resonator and the cavity for split-ring resonators.

Perspectives and future work

1. The finite elements approach developed and presented in Chapter 3 has been tested and validated for general layers. However, in Chapter 4, dispersion relations are only studied for the case of equivalent fluid; future work thus include an extension of the study of dispersion relation of waves in metaporous structures, by assuming poroelastic layers and solid inclusions, leading to a much larger of modes to analyse due to the presence of the modes linked to the elastic phase of the materials modeled with the full Biot theory. As observed for homogeneous layers, this leads to a much larger number of modes to be solved. While the numerical implementation does not consistute an issue as the basis of it was validated in Chapter 3, the analysis of the results is expected to be quite complex. investigate more deeply the physics, energy conversion mechanisms and such

2. An important perspective for the metaporous structure dispersion relation computation presented in Chapter 4 would involve the development of a more robust method not relying on a root-finding procedure. Some time has been spent during the development of the method to try to write the equations in the form of an eigenvalue problem with k_{1q} as the eigenvalue , although this was not achieve due to the challenges of obtaining such a formulation with the way the coupling between the surrounding medium and the structure is done. The finite elements matrices obtained in Eq. ref the correct eq depend on both on a term in $e^{ik_{1q}d}$ and an integral on x_1 with a $e^{ik_{1q}}x_1$ term. An appropriate change of variable or a different formulation for the coupling might provide a way to achieve the writing of an eigenvalue problem. Note that this has been explained briefly in Section. ref annexe PRB and shows that the eigenvalue problem is easily written when not accounting for the surrounding medium.

Methods like contour integral evaluation by using the Beyn method or the FEAST eigenvalue solver.[guttel2017, liu2025] Although these iterative procedures have the advantage of not requiring initial guesses as initialisation, the contour integrals are still computed around a given value, thus still requiring some sort of initialization. Additionally, the quality of the matrix conditioning should be investigated as well due to the amount of sensitive operations required by these methods (e.g. the successive SVD should be computed with sufficient precision).

3. A study the excitability of the modes could be conducted by using a numerical model done with COMSOL & a line source excitation properly tuned according to the dispersion relation modes to confirm the form of the solutions of the modes in finite structures and a more *realistic* configuration and further link the dispersion relation results to an actual observable behaviour of the structures by gaining insights on the excitability of the modes. This is an important aspect as there are many dissipation mechanisms involved, leading to very attenuated modes: seeing if they can be excited or if their existence actually affects the wave propagation in any way.

near field nature of leaky modes, wannier functions, extension to 3d w/ out of plane modes Simulation results in these situations could allow for further experimental validation for the configurations of interest. One could imagine a porous sample manufactured with rigid cylindrical inclusions. There would be challenges to this, including the precision required to engineer precisely the radius of the cylinders. Measurements need to take into account the nature of the source needed to excite the variety of these modes. A post-process of the frequency responses using SLaTCoW or some other method allowing for the retrieval of complex wavenumbers. This approach is similar to that applied in Section.**??**.

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